

Total No. of Questions : 4]

SEAT No. :

PD-392

[Total No. of Pages : 2

[6411]-185
B.E. (AI&DS)
Distributed Computing (Insem.)
(2019 Pattern) (Semester - VIII) (417531)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of electronic pocket calculator is allowed*

- Q1)** a) What are the key features and characteristics of distributed pervasive computing? **[5]**
- b) How would you analyze and illustrate the intricate relationship between middleware and the three-tier architecture through a detailed diagram, demonstrating their interconnected functionalities and roles within a system? **[6]**
- c) Can you summarize the key components and objectives of Intelligent Transportation Systems (ITS) in a brief overview? **[4]**

OR

- Q2)** a) Can you enumerate the various types of distributed systems? **[5]**
- b) What are the primary challenges related to data storage and retrieval in distributed computing environments, and how do they influence the overall performance and efficiency of the system? **[6]**
- c) Can you summarize the key characteristics and distinguishing features of different distributed system models in a concise manner? **[4]**

- Q3)** a) “Why is there a necessity for the Google File System (GFS), and could you elaborate on its workings in detail? [4]
- b) Explain Any Two from below: [6]
- 1) Eager Replication,
 - 2) Lazy Replication
 - 3) Quorum based Replication.
- c) Discuss two consistency models in detail. Why is consistency important? Justify your answer. [5]

OR

- Q4)** a) Differentiate between Amazon web services, Azure Cloud and Google cloud Platform? [4]
- b) Explain Message Broker and Stream Processing in detail ? [6]
- c) Explain Distributed hash table, distributed inverted Indexing? [5]

